PHYSICAL SCIENCE
CHEMISTRY CYCLE TEST

Grade 10

24 March 2026

Time: 42 mins

46 marks +3.

ASSESSOR: K. Bolton

MODERATORS: G. Main and T. Kirsten

INSTRUCTIONS:

1. This question paper consists of 7 pages, and a data sheet. Please check your question paper is complete.
2. All questions are to be answered on the question paper.

NAME: _____ MEMO _____ **SCIENCE TEACHER:** _____

QUESTION 1 MULTIPLE CHOICE

Answer these questions on the multiple-choice grid below by placing a cross over the letter of your choice.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D

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1.1 Element *X* is in group 2, and element *Y* in group 7, of the periodic table. Which ions will be present in the compound formed when *X* and *Y* react together?

A X^{+} and Y^{-}

B X^{2+} and Y^{-}

C X^{+} and Y^{2-}

D X^{2-} and Y^{+}

1.2 Based on electronegativity values, which bond is the most polar?

A B—C

B C—O

C N—O

D O—F

1.3 The number of electrons in the valence shell of elements X and Y are 5 and 7 respectively. What is the formula and type of compound formed from these elements?

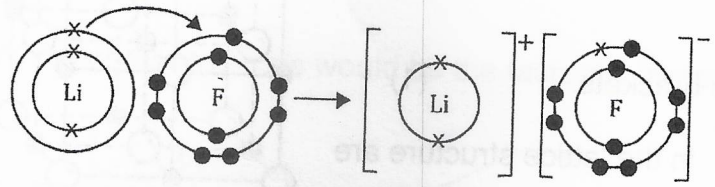
A Covalent, XY_3
B Ionic, XY_3
C Covalent, X_3Y
D Ionic, X_3Y

[2X3=6]

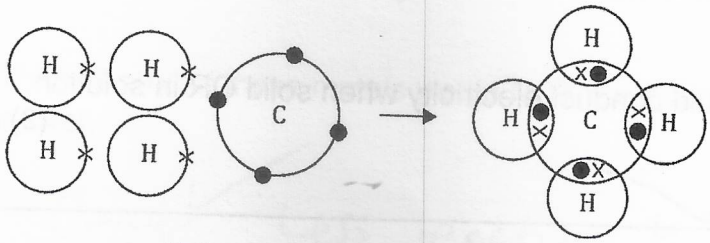
ESTION 2

The diagrams below illustrate THREE different types of bonding.

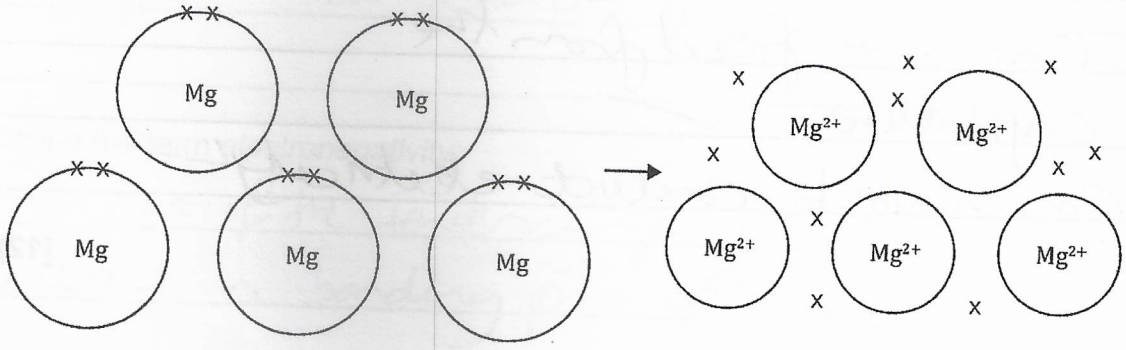
A.



B.



C.



2.1 Complete the sentences comparing the electron distribution in different substances:

(a) In covalent compounds, the bonding electrons are shared ✓.

(b) In ionic compounds, electrons are transferred ✓.

(c) In metals, the electrons are delocalised ✓.

2.2 In each case, **CIRCLE** the correct word between the brackets: (3)

(a) The type of bonding in figure A is **METALLIC/IONIC/COVALENT.**

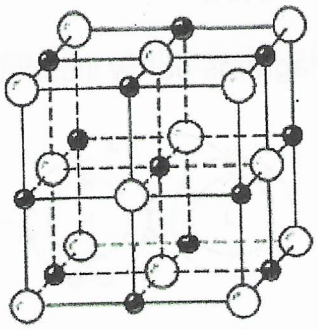
(b) The type of bonding in figure B is **METALLIC/IONIC/COVALENT.**

(c) The type of bonding in figure C is **METALLIC/IONIC/COVALENT.** (3)

2.3 Why are metals good conductors of electricity? (2)

Sea of delocalised electrons ✓ which can move ✓

2.4 The diagram illustrates the crystal lattice structure of solid sodium iodide.



CIRCLE the correct word within the brackets. (1)

The particles represented by  in the lattice structure are

(ATOMS / MOLECULES / IONS).

2.4.1 Explain whether this substance would conduct electricity when solid OR in solution. (3)

In solution ✓
Ions are freed from the
crystalise ✓
Can move to conduct electricity ✓

[12]

ESTION 3

Equal sized pieces of lithium, magnesium and copper are placed in separate test tubes containing hydrochloric acid.

3.1 Which of these metals would be the fastest to react with the acid? (1)

Lithium ✓

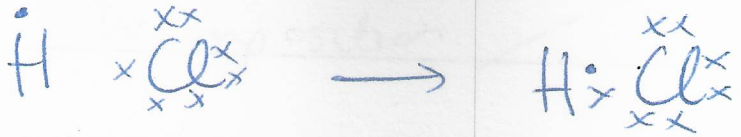
3.2 Which, if any, of these metals would NOT react with the acid? Give a reason for your answer. (2)

Copper ✓ Less reactive than hydrogen ✓

3.3 Define the term *electronegativity*. (2)

measure of the tendency ✓ of an atom to attract a bonding pair of electrons ✓

3.4 Draw a Lewis diagram to show the formation of a bond in a molecule of hydrochloric acid. (3)



3.5 Using electronegativity numbers, state the SPECIFIC type of bonding in hydrochloric acid. (3)

H - 2,1 ΔEN = 0,9 ✓

Cl - 3,0 ∴ POLAR COVALENT BOND ✓ ✓

3.6 Magnesium reacts with hydrochloric acid to form a salt and a gas.

3.6.1 Write a balanced chemical equation to show the reaction between magnesium and hydrochloric acid. (3) +1

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

3.6.2 NAME the type of reaction between magnesium and hydrochloric acid? (1)

Displacement

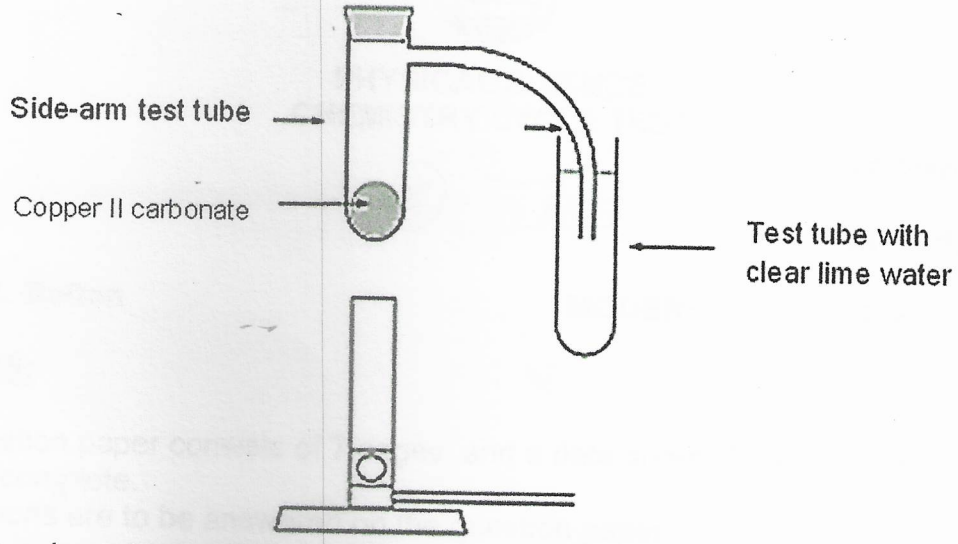
3.6.3 Describe how you would test for the gaseous product? (2)

Let splint out with a POP

[17] +1

ESTION 4

Copper (II) carbonate is heated, and the products are a solid and a gas. The gas that is released, is bubbled through clear limewater.



4.1 State the colour change that would be observed in the side arm test-tube. (2)

Turquoise/ blue-green → black.

4.2 NAME the TYPE of reaction. (1)

Decomposition

4.3 NAME the solid product produced from this reaction. (2)

Copper (II) oxide

4.4 NAME the gas product and give the result of the limewater test. (2)

Carbon dioxide Turns milky

4.5 **Lead (II) nitrate + ammonium hydroxide → ammonium nitrate + lead (II) hydroxide**

4.5.1 Rewrite the equation above in its balanced form, using formulae. (5)

$Pb(NO_3)_2 + 2NH_4OH \rightarrow 2NH_4NO_3 + Pb(OH)_2$ bal.

4.5.2 NAME the TYPE of reaction. (1)

Partner / Ion exchange.

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